

COP2800C Module 3 Graded Programming Assignment

In this assignment, we will extend the functionality of the Palmer Penguins program by introducing branching logic using if-else and switch statements.

Instructions

1. Start with the Module 2 Solution (included in Module 3 in Canvas); rename the class as `PalmerPenguinsM3` and save the file as `PalmerPenguinsM3.java` (remember that a public class must always be stored in a file with the same name as the class).

- Be sure to update the 4-line ID header at the top of the file, reflecting the new file name, submission date, and summary.

2. Add the new functionality:

- Start with an analysis using if-else which determines which penguin species is the most populous.
 - Compare the population counts (`NUM_CHINSTRAP`, `NUM_GENTOO`, `NUM_ADELIE`) to determine which species has the highest population.
 - Handle ties by printing a message indicating if there's a tie for the most populous species.

Example Output:

```
Branching Analysis:
Gentoo penguins are the most populous.
```

Example Output if there is a tie:

```
Branching Analysis:
There is a tie between species for the most populous.
```

- Be sure to use correct indentation as shown here:

```
if (NUM_CHINSTRAP > NUM_GENTOO && NUM_CHINSTRAP > NUM_ADELIE) {
    System.out.println(SP_CHINSTRAP +
        " penguins are the most populous.");
} else if ... {
    ...
} ...
// you will need an unconditional else at the end for the tie
```

- Now add an analysis using a switch statement which displays the population count and percentage for a single species based on the value of a variable. Use a String variable (e.g., chosenSpecies) to represent the species being analyzed:

```
// Change this to analyze a different species  
String chosenSpecies = SP_GENTOO;
```

- Include a case statement **for each species** to display the values (but remember that only one species will be displayed).
- Be sure to use correct indentation as shown here:

```
switch (chosenSpecies) {  
    case SP_CHINSTRAP:  
        System.out.printf("%s: %d (%.2f%%)\n",  
            SP_CHINSTRAP, NUM_CHINSTRAP,  
            ((double) NUM_CHINSTRAP / totalPenguins * 100));  
        break;  
    case SP_GENTOO:  
        ...  
    // don't forget your default case and a closing }
```

Example Output:

```
Switch Statement Analysis:  
Gentoo: 123 (35.96%)
```

Example Output if the species is not included in a case statement (the **default** case):

```
Switch Statement Analysis:  
Species not recognized.
```

- Be sure to maintain clean, readable code which follows our course coding conventions:
 - Use 3-4 spaces for each level of indentation; do not use tabs.
 - Separate logical sections with a blank line.
 - Keep lines under 80 characters long.
 - Include descriptive comments.

4. Test Your Program

- Run your program to verify the output. It should:
 - Introduce the penguin species and their populations (retain this feature from Module 2).

- Perform the branching analysis to identify the most populous species.
 - Perform the switch-case analysis for a specific species, displaying its population and percentage.
- Modify the data (constant values and chosenSpecies variable) to validate your "tie" branch and your invalid species case.
- Recompile and execute for different test values, do not add extra logic for testing (e.g., loops). Do not include language features that we have not covered in class yet.

5. Screenshot Your Output

- Take screenshots of your program's output in the "Run I/O" pane of your IDE (e.g., jGRASP). Do not screenshot source code.
- Your screenshots should **minimally** include test results of the analyses for various values, including results for testing the tie branch and the invalid species case.
- The introductory output from Module 2 does not need to be in your screenshots but the code should be retained.

Submission

1. Submit the PalmerPenguinsM3.java file to your GitHub repository.
 - Do **not** include .class files.
 - Ensure your code adheres to our coding conventions.
2. Upload Your Screenshots to the repository.

Sample Output

The output of your program should be similar to the following, although your specific values may vary based on your selection of species and experiments with the species totals (percentages may vary due to rounding):

```

Introducing the Palmer Penguins:
    Chinstrap!
    Gentoo!
and last but not least...
    Adelie!

There are a total of 3 penguin species in this dataset.
There are a total of 342 penguins in the dataset.
Chinstrap: 68 (19.88%)
Gentoo: 123 (35.96%)
Adelie: 151 (44.16%)

Branching Analysis:
Gentoo penguins are the most populous.

Individual Species Switch Statement Analysis:

```

Gentoo: 123 (35.96%)

Submission

Submit your .java file and screenshots to the GitHub repo.